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Your F5 Platform Integration Partner

3PLZ Technical Proposal for CVS Health

CE-Centric | XC Native Proxy | Per-Tenant Isolation | mTLS + WAF + DNS + SSH

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1. Executive Summary

1.1 What This Platform Does

3PLZ is a secure network access control gateway that brokers third-party connectivity to CVS systems. It terminates IPSec tunnels from external partners (Cisco ASA, Palo Alto, SD-WAN), enforces mutual TLS (mTLS) on application traffic, applies WAF/DDoS protection per tenant, provides DNS mediation, and hosts a lightweight SSH bastion with multi-factor authentication. CVS owns the data. WTIT hosts and operates the infrastructure. Deployment is 3 months to a 5-tenant POC, scaling to 50+ tenants within 12 months.

1.2 What This Platform Does NOT Do

This platform is NOT designed for HIPAA workloads by default. It is NOT a Business Associate, does NOT process PHI, and does NOT replace CVS's own compliance obligations. Where PHI or regulated data flows exist, the platform provides L4 pass-through (encrypted tunnel, no inspection) and CVS retains full data classification responsibility.

1.3 Business Value Proposition

Single Landing Zone Replaces Multi-Point Tunnels: Today, CVS maintains separate network connections per third-party partner, each with its own firewall rules, VPN config, and monitoring. 3PLZ consolidates all third-party ingress into one governed gateway, reducing operational surface area and enabling centralized policy enforcement.

Enhanced Security Without Partner Changes: Full inspection (mTLS, WAF, API visibility) is applied at the gateway layer. Partners connect via standard IPSec VPN and see no change to their existing infrastructure or workflows. Security upgrades happen at the platform level, not per-partner.

Scales from 5 Tenants to 50+: The architecture is designed to grow cleanly. MVP delivers 5 production tenants in 3 months. Scaling beyond 25 tenants per cluster adds new CE clusters via Terraform—no re-architecture required.

1.4 Investment & Timeline

The 3PLZ POC is a fixed-fee engagement. WTIT hosts all infrastructure and covers F5 XC licensing through the WTIT MSP program.

Item	Value
Total POC Cost	$290,000
POC Duration	3 months from project kickoff
Tenants Onboarded (POC)	Up to 5
POC Is Production-Grade	Yes; no rip-and-replace for production scale
Path to 50+ Tenants	Defined in Phase 2-3 roadmap
POC Billing Model	Fixed fee

Infrastructure	WTIT Cloud.Red Platform – includes hosted AWS environment, and F5 XC

1.5 Key Architecture Highlights

The Cloud.Red platform is built on AWS and F5 Distributed Cloud with WTIT-managed Customer Edge nodes. The following table summarizes the core architectural components for CVS' deployment:

Component	Implementation
CE Deployment	WTIT-hosted CE farm (AWS), managed via XC SaaS console (SLO interface only)
IPSec Termination	Routed mode (VTI) via External Connector; PSK authentication only
mTLS + L7 Inspection	XC HTTP Load Balancers: mutual TLS, WAF, and optional API visibility per namespace
L4 Pass-Through	XC TCP/UDP Load Balancers: connection-level only (no payload inspection)
Per-Tenant Isolation	XC namespaces + Network Policies + Segments
Certificate Lifecycles	Infrastructure certs (1-2yr, platform identity) + Server certs (90-day, LB TLS) + Client certs (90-day POC / 7-day prod, mTLS zero-trust)
DNS Mediation	CoreDNS containers (Fargate/Docker) fronted by XC TCP/UDP LBs
SSH Bastion	OpenSSH + TOTP 2FA container (Fargate/Docker), per-tenant, behind XC TCP LB

1.6 Engagement Model

CVS Health is the customer and data owner. World Tech IT (WTIT) is the delivery partner – Utilizing their Cloud.Red platform with custom automations built for CVS, they are the operator, and managed service provider. WTIT hosts and manages all infrastructure including F5 XC, AWS compute, networking, and storage. CVS pays a single fixed fee for the POC engagement, and if success criteria is met will move forward with the ongoing managed service.

2. Solution Architecture

2.1 Three CE Classes

The architecture supports three Customer Edge deployment models, each serving a different connectivity role within the 3PLZ platform:

CE Class	Location	Use Case	Sizing
WTIT-Hosted CE Farm	AWS (REGION TBD), WTIT-managed	Primary ingress; IPSec termination + proxy for all partner traffic	M5.2xlarge (POC and production 3-node HA)
CVS Border CE	CVS data center	Connection broker between CVS and 3PLZ infra	c5.2xlarge or equivalent
Partner-Side CE (Optional future Add-On and not in scope at this time)	Partner premises	When partner requires local termination (Phase 2+)	Sized per partner requirements

2.2 IPSec Specification

Each tenant's IPSec tunnel lands on a dedicated pseudo Site Local Inside (SLI) interface, creating isolated forwarding domains per tenant. This ensures complete traffic separation at the network layer.

Parameter	Value
Mode	Routed (VTI) — only supported mode
Authentication	Pre-Shared Key (PSK) — only supported authentication method
Tunnel Interface	Virtual Tunnel Interface (VTI) per tenant, landed on dedicated Segment (Isolation Wrapper)
Encryption	AES-256-GCM, SHA-512, DH Group 19
Tunnel Capacity	Up to 25mbps per tunnel
WAF & API Visibility	Combined up to 10,000 hits
Routing	BGP preferred over VTI; static routes as fallback
MTU	1400-1436 recommended (PMTU discovery NOT supported)
IKE Version	IKEv1/IKEv2 based on peer device

Each 3-node CE cluster supports up to 25 concurrent VPN peers.

2.3 Network Segments

WTIT manages CEs exclusively through the F5 XC SaaS console via the SLO interface. SLI interfaces are NOT used for management, they serve only as isolation wrappers to land IPSec tunnels into dedicated forwarding domains.

Segment	Function	Traffic Types
Site Local Outside (SLO)	Internet-facing; connects to XC Regional Edges for management and control plane	CE management, control plane, telemetry, health checks
Segment (Isolation Wrapper)	Dedicated interface per tenant to land IPSec tunnel VTI	Tenant IPSec tunnel traffic ONLY; no management traffic

2.4 Load Balancer Architecture

All listeners in F5 XC are created via Load Balancers. There is no separate ‘proxy’ concept—everything is an Load Balancer (LB). The three LB types serve different inspection levels:

LB Type	Protocol	Use Case	Tenant Isolation
HTTP Load Balancer	HTTPS (L7)	Web/API traffic: mTLS termination, WAF, bot protection, API visibility (future optional schema enforcement – not in current scope)	Per-tenant namespace; dedicated LB config
TCP/UDP Load Balancer	TCP or UDP (L4)	Non-HTTP protocols, SSH bastion, DNS mediation	Per-tenant namespace; port-based separation
Forward Proxy Policy	HTTPS (L7 Light) or TCP/UDP (L4 Light)	As needed (One-Off)	Segment to Segment isolation

3. Certificate Strategy (Three Lifecycles)

The platform manages three certificate lifecycles, each with its own threat model, rotation cadence, and automation requirements. This separation is deliberate: infrastructure certs protect platform identity, server certs protect LB TLS identity, and client certs enforce zero-trust access control.

3.1 Infrastructure Certificates (Platform Identity)

These certificates provide identity and authentication for platform infrastructure components: CE enrollment, internal service communication, and monitoring.

Parameter	Value
Use Case	Platform infrastructure identity: CE enrollment, internal service auth, monitoring
Validity	1-2 years (configurable)
Key Algorithm	ECDSA P-256 or RSA 2048
SAN / CN	platform.3plz.cvs.internal, ce-{id}.3plz.cvs.internal
Revocation	CRL published by step-ca, refreshed every 12 hours
Renewal	Cloud.Red alerts 60 days before expiry; renewal is automated
Delivery	Pushed to platform components by Cloud.Red automation
Key Custody	Root CA offline cold storage; Intermediate CA key held in AWS KMS (ECC P-256)

3.2 Server Certificates (XC HTTP Load Balancer Identity)

These certificates provide TLS identity for the XC HTTP Load Balancers fronting each tenant's web/API services. They are NOT a zero-trust mechanism—they prove the LB's identity to connecting clients.

Parameter	Value
Use Case	TLS server identity on XC HTTP Load Balancer
Validity	90 days (POC and production)
Key Algorithm	ECDSA P-256
SAN / CN	tenant-{id}.apps.3plz.cvs.internal
Revocation	CRL (12-hour refresh); revoke and re-issue if key compromise suspected
Renewal	Cloud.Red automation, 7 days before expiry
Delivery	Cloud.Red pushes cert to XC certificate store via API
Hot Reload	XC picks up new cert in-place, zero-downtime

3.3 Client Certificates (mTLS Zero-Trust Enforcement)

Client certificates are the zero-trust enforcement mechanism. They prove the connecting entity's identity to the XC HTTP LB. Without a valid client cert, the connection is rejected at the TLS handshake—before any application logic executes.

Parameter	POC (Phase 1 + 1.5)	Production (Phase 2+)
Use Case	mTLS client identity	mTLS client identity
Validity	90 days	7 days
Key Algorithm	ECDSA P-256	ECDSA P-256
SAN / CN	tenant-{id}.client.3plz.cvs.internal	tenant-{id}.client.3plz.cvs.internal
Revocation	CRL (12-hour refresh)	CRL + OCSP (when deployed)
Renewal	Cloud.Red scheduled job; manual reissuance if needed	Fully automated via operator portal + Cloud.Red API
Rationale	No frontend for reissuance; 90-day lifetime avoids operational friction during POC validation	7-day lifetime = core zero-trust posture; compromised cert has max 7-day window

3.4 Tenant Isolation via Provisioner Configuration

Each tenant receives three dedicated provisioners in step-ca: one for infrastructure certs, one for server certs, and one for client certs. SAN constraints enforce strict tenant boundaries.

Provisioner	Max Duration	SAN Allow	SAN Deny
tenant-{id}-infra	730 days	tenant-{id}.3plz.cvs.internal, ce-{id}.3plz.cvs.internal	*.cvs.com, *.cvs.internal, other tenants
tenant-{id}-server	90 days	tenant-{id}.apps.3plz.cvs.internal	*.cvs.com, *.cvs.internal, other tenants
tenant-{id}-client	90 days (POC) / 7 days (prod)	tenant-{id}.client.3plz.cvs.internal	*.cvs.com, *.cvs.internal, other tenants

Cloud.Red manages renewal schedules per provisioner type. Server certs renew 7 days before expiry. Client certs renew based on lifecycle policy (POC: 7 days before expiry; production: 1 day before expiry for 7-day certs). Infrastructure certs renew 60 days before expiry.

3.5 PKI Key Custody

AWS KMS integration is confirmed feasible via step-ca's native awskms provider (using kms:Sign and kms:GetPublicKey operations). This provides hardware-grade key protection without dedicated hardware appliances.

Aspect	Implementation
Root CA Key	Offline cold storage; used once to sign intermediate CA
Intermediate CA Key	AWS KMS asymmetric key (ECC P-256); all signing operations performed by KMS
KMS Region	Same AWS region as CE deployment
Credential Management	IAM instance role on step-ca EC2; no credential files on disk

OCSP responder is deferred to Phase 2. CRL is sufficient for POC and Phase 1. Estimated CRL refresh interval: 12 hours, distributed to XC via Cloud.Red automation.

4. Traffic Inspection Model

4.1 Design Principle: Default vs. Exception

The platform's value is not connectivity alone, but enhanced security and contextual awareness of third-party traffic. Full L7 inspection is the default posture. L4 pass-through is the exception, used only where compliance or regulation explicitly prohibits inspection.

4.2 DEFAULT: Full Inspection (Where Compliance Allows)

By default, all traffic passing through 3PLZ is subject to full Layer 7 inspection via XC HTTP Load Balancers. This includes:

Control Point	Mechanism	Capability
mTLS Termination	SSL/TLS decryption on XC HTTP LB	Validate client certificate CN → user id
WAF (Web App Firewall)	OWASP CRS baseline per namespace	SQL injection, XSS, path traversal blocks
API Visibility	API Visibility - schema Discovery and Behavioral Analysis	Schema Compliance, Outlier Detection, API Vulnerability Detection
Anomaly Detection	Pending F5 investigation	Capabilities TBD; pending Joel Moses investigation
Session Logging	HTTP request/response capture	Headers, status codes, latency, response size
Identity	Certificate CN or HTTP header	User ascribed via mTLS cert or custom header

4.3 EXCEPTION: L4 Pass-Through for Compliance-Restricted Flows

If a specific service is identified as PHI, PCI-DSS, or other compliance-regulated, it is routed through a TCP/UDP Load Balancer instead. This provides:

Aspect	Behavior
SSL/TLS Termination	No (traffic passes encrypted end-to-end)
Payload Inspection	No (no transformation, modification, or decryption)
SNAT	Yes (bidirectional: partner IP masqueraded as CE SNAT IP; CVS sees CE only)
Connection Logging	Only (source IP, dest IP, port, byte count)
Error Logging	Only (no command or content capture)
Certificate Use	Authentication/identity only, not decryption
Audit Trail	Proves WTIT/F5 never accessed plaintext data

5. HIPAA/Compliance Position Statement

5.1 Explicit Non-Compliance Statement

This platform is NOT designed for HIPAA workloads by default. World Tech IT makes no representations regarding HIPAA compliance, does not sign BAAs for the 3PLZ platform, and does not accept responsibility for protected health information (PHI) transiting the platform.

The platform is NOT:

- A HIPAA-compliant data processing system
- A healthcare data enclave
- Covered under HIPAA Title II by default
- Subject to Business Associate Agreement (BAA) requirements by default
- Designed to enforce HIPAA-specific audit trails, encryption standards, or access controls

5.2 Regulatory-Sensitive Flows: L4 Pass-Through Handling

If CVS identifies a PHI flow during service onboarding, the platform handles it via L4 pass-through: encrypted end-to-end, no inspection, connection-level logging only. This ensures WTIT Cloud.Red never has access to plaintext PHI data.

CVS Health retains full responsibility for: data classification accuracy, partner BAA management, HIPAA Security Rule compliance, and audit trail retention.

5.3 Responsibility Matrix

The following matrix defines accountability for compliance-related controls.

Control	Responsible Party	Evidence
Data classification (PHI vs. non-PHI per service)	CVS Health	Onboarding interview documentation
Selection of Inspection Mode (Full vs. L4 Pass-Through)	CVS Health	Service registration in Cloud.Red
HIPAA Security Rule compliance	CVS Health	BAA with backend data owners
Encryption in transit	CVS Health (negotiates with partners) + WTIT (enforces tunnel)	IPSec tunnel + mTLS (if applicable) + audit logs
Access control enforcement	CVS Health + WTIT	Namespace policies + audit logs
Audit trail retention and legal holds	CVS Health	CloudWatch logs or SIEM integration

6. Cloud.Red Orchestration (WTIT Automation Platform)

Cloud.Red is WTIT's tenant lifecycle automation platform built on F5 XC APIs and Terraform. It manages the entire lifecycle of each tenant from onboarding through offboarding, including certificate management, DNS provisioning, and audit logging.

Core capabilities:

- Tenant onboarding API (create, modify, suspend, offboard)
- Automated IPSec config generation (PSK configuration package)
- step-ca provisioner management and certificate renewal
- Terraform-driven XC resource creation (namespaces, LBs, policies)
- Per-tenant Corefile generation for DNS mediation
- Audit logging to Splunk (access events, policy changes, cert rotations)
- Scheduled cert renewal jobs (server certs every 83 days, client certs per lifecycle policy)

7. Phased Roadmap

7.1 Phase 1: Foundation (Months 1-3) — POC — Full 5-Tenant Delivery

- WTIT-hosted CE farm (1 node) live on AWS
- Up to 5 tenants connected (partner IPSec tunnels via External Connector)
- XC HTTP LBs with mTLS enforced, WAF active per tenant namespace
- step-ca operational (all three cert lifecycles tested)
- DNS mediation operational (public FQDN → private landing zone IP resolution)
- SSH bastion MVP deployed (OpenSSH container + TOTP 2FA, per-tenant, behind XC TCP LB)
- Security review submitted

7.1.1 DNS Mediation (Phase 1)

CoreDNS runs as lightweight containers with per-tenant zone files. No AWS managed DNS services (expensive managed DNS services). Zone management is driven by Cloud.Red on each tenant onboarding.

Component	Deliverable	Exit Criteria
CoreDNS Containers	CoreDNS deployed on Fargate (or Docker), per-tenant zone files	Public FQDN → private tenant IP resolution verified
DNS Load Balancing	F5 XC TCP/UDP LBs front CoreDNS containers (port 53)	Queries route correctly; <10ms latency
Zone Management	Cloud.Red maintains per-tenant CoreDNS zone files	New tenant DNS provisioned within 5 minutes of onboarding

7.1.2 SSH Bastion (Phase 1)

Phase 1 SSH bastion is MVP-only: OpenSSH + TOTP 2FA, no Vault SSH Secrets Engine. Session recording is deferred to Phase 1.5.

Component	Deliverable	Exit Criteria
SSH Proxy	OpenSSH container (Alpine) + PAM (TOTP 2FA)	SSH via XC TCP LB succeeds; 2FA works
Container	Per-tenant container behind XC TCP LB (port 22)	Image <80MB; cold start <5s
Backend Auth	Static SSH keys (MVP)	User auth via PAM; backend connectivity verified
Session Audit	Basic syslog forwarding to SIEM	Login events logged; searchable

7.2 Phase 1.5: SSH Enhanced Logging — POC Continuation & Visibility (Bolt-On)

Phase 1.5 is a bolt-on initiative that extends Phase 1 SSH bastion capabilities with session recording and enhanced audit logging. It runs concurrently with late Phase 1 delivery.

Component	Deliverable	Timeline
Session Recording	tlog or equivalent for terminal I/O replay + forensic audit	Weeks 12-15
SSHPiper Evaluation	Architecture review; PoC if warranted. Note: limited to password + public key auth	Weeks 12-15

7.3 Phase 2: Operational MVP (Months 4-6)

Phase 2 is 100% operationalization: it transforms POC engineering workflows into production-grade automated services. No new features are introduced—Phase 2 hardens and automates everything delivered in Phase 1 + 1.5.

Core Deliverables:

- Tenant Lifecycle APIs (RESTful automation for Provisioning, Offboarding, User Management)
- Certificate Automation (server + client certs, automated renewal, audit logging)
- Operator Portal v1 (web UI for complete tenant lifecycle, user management, cert status)
- DNS Mediation Automation (per-tenant CoreDNS pod, service catalog-driven Corefile generation, Cloud.Red integration)
- Monitoring & Alerting (certificate expiry, IPSec tunnel health, gateway resource utilization, Splunk/CloudWatch)
- Operational Runbooks (onboarding, offboarding, certificate rotation, failure recovery)
- Vault SSH Integration (Vault SSH Secrets Engine for ephemeral SSH credentials)

Target: End of Month 6

7.4 Phase 3: Hardening (Months 7-12)

- 10+ tenants total, 50+ tenant load test
- Infrastructure scaling (multi-AZ, multi-region optional)
- step-ca Kubernetes HA (StatefulSet, failover)
- OCSP responder deployed
- Self-service tenant portal (basic status, cert info, logs)

8. Pricing

8.1 POC Investment (Phase 1 + Phase 1.5)

CVS Health pays a single fixed fee. WTIT Cloud.Red platform includes all infrastructure hosting and F5 XC licensing through the WTIT MSP program for 5 Tenants to prove success criteria. Once success criteria is met, the cluster will have capacity to cut over an additional 20 b2b partners – which is included in the production Cloud.Red 3PLZ Base Platform.

Item	Description	Cost
Cloud.Red 3PLZ POC	3 Month POC to build the initial production infrastructure that will be used to prove the defined success criteria. The POC will include all items described in this document including; architecture, design, implementation, testing, and cut- over support for 5 sites. To be completed within 3 months.	$290,000
Success Criteria	• Successfully cut over 5 sites • Successful proxy and inspection of HTTP/S traffic • Successful proxy and connection brokering for SSH	

8.2 Post-POC Production Run Rate (Estimated)

Once success criteria is met, the following estimates reflect production-grade pricing for ongoing managed service delivery. These are projections based on the current scope and information we've been provided, if the scope changes costing will be finalized post-POC based on new requirements. Post-POC production costs are estimated below based on the base platform + tenant pack fees.

8.2.1 Platform Costs Base + Per CE Cluster & Supporting infrastructure.

Component	Description	Annual Cost
Cloud.Red 3PLZ Base Platform	Managed Service Includes all base infrastructure and endpoints to pragmatically interface with automation for site stand up. Fixed cost for Cloud.Red & supporting infrastructure to support up to 25 tenants. Includes: • Tenant onboarding • Certificate lifecycle management • All backend services • DNS zone maintenance • Monitoring & Alerting • 24 x 7 Managed Support *Note Excludes front end GUI interface*	$500,000
Additional Tenant Cluster	Fixed cost for Cloud.Red & supporting infrastructure to support up to 25 tenants. Includes: • Tenant onboarding • Certificate lifecycle management • All backend services defined in this doc • DNS zone maintenance • Monitoring & Alerting • 24 x 7 Managed Support *Note Excludes front end GUI interface*	$350,000

9. Open Questions (Unknowns)

9.1 Critical Path (Phase 1, Week 1-2)

These questions must be resolved during Phase 1 kickoff (Weeks 1-2) to validate the architecture and unblock implementation.

Question	Why It Matters	Who Resolves	When
External Connector Segment Routing	Must validate on both AWS-hosted CEs and partner-side (optional) CEs to match CVS criteria	WTIT	Phase 1 Week 1-2

CVS DC compute readiness	Do we have 8 vCPUs + 32GB RAM + 80GB disk x3 available for border CE cluster ?	CVS Infra + WTIT	Phase 1 Week 1
XC HTTP LB gRPC support	Are bidirectional streaming protocols (gRPC) required on HTTP LB?	CVS + WTIT	Phase 1 Week 2
step-ca CRL distribution	How do partner devices validate CRL for platform infrastructure certs?	WTIT PKI team	Phase 1 Week 3

9.2 Phase 1 Scoping (Weeks 3-4)

These questions scope Phase 1 deliverables and clarify partner/CVS infrastructure dependencies.

Question	Why It Matters	Who Resolves	When
Anomaly detection capabilities	Feature scope is TBD pending investigation with XC engineering	WTIT (pending Joel Moses investigation)	Phase 1 end
SSH Phase 1.5 scope	What session recording / SSH enhanced logging capabilities are achievable?	WTIT + CVS	Phase 1 Week 4
DNS mediation implementation	Which lightweight DNS approach fits AWS/Fargate POC constraints?	WTIT + CVS (Functionality Check)	Phase 1 Week 3

9.3 Phase 2 Deferral Questions

These questions do not block Phase 1 delivery and can be deferred safely into Phase 2 planning.

Question	Why It Matters	When
OCSP responder deployment	Reduces revocation latency but adds ops overhead; CRL sufficient for POC	Phase 2 planning
CoreDNS scaling model	Per-tenant pods simplest for MVP; shared pods need evaluation at scale	Phase 2 review

10. Architectural Decisions10.2 Decide Later

These decisions can safely be deferred without blocking Phase 1 delivery.

Decision	When	Why Defer	Rationale
Multi-region CE topology	Phase 3	Single region sufficient for 5-10 tenants	Geographic redundancy not needed until scale
OCSP vs. CRL for infra cert revocation	Phase 2	CRL sufficient for POC	OCSP adds ops overhead prematurely
CoreDNS shared vs. per-tenant pods	Phase 2	Per-tenant simplest for MVP	Evaluate shared model at 10+ tenants
Network Engine migration timing	Phase 3	Existing infra sufficient	Only relevant if border CE deployed

Conclusion

3PLZ is a production-grade, scalable platform that brokers secure third-party network access for CVS Health. The Cloud.Red platform is built on AWS and F5 Distributed Cloud with WTIT-managed Customer edge nodes, providing centralized policy enforcement, per-tenant isolation, and full traffic inspection (where compliance allows) without requiring partner infrastructure changes.

Key success factors: (1) resolving open questions in Phase 1 Week 1-4, (2) validating External Connector segment routing on real hardware, (3) establishing certificate lifecycle automation early, and (4) proving the cost model at 5 tenants before committing to scale. The 3-month POC is designed to deliver production-grade infrastructure that requires no rip-and-replace to scale to 50+ tenants.

The platform is not a HIPAA solution by default. CVS retains full control over data classification, partner BAA management, and compliance obligations. WTIT provides the infrastructure, automation, and managed service layer. The partnership is designed for trust, transparency, and technical excellence.

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